

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 23: Graphical Representation (Histograms and Ogives)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Mode from a histogram

1. Draw a histogram for the marks of 164 students given below, and use it to find the mode.

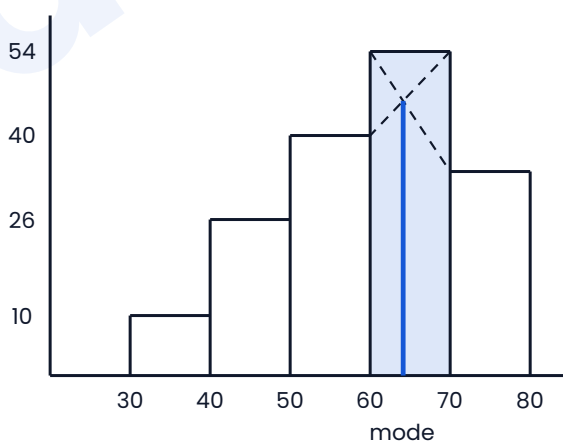
ICSE 2025, 3 marks

Marks	30–40	40–50	50–60	60–70	70–80
Number of students	10	26	40	54	34

SOLUTION

Draw a rectangle on each class with height equal to its frequency. The tallest rectangle (60–70) is the modal class.

Join the top corners of the tallest rectangle to the nearer top corners of the rectangles on either side, as shown. From the point where the two lines cross, drop a perpendicular to the marks axis.



The perpendicular meets the axis at about 64.

Check with the formula: $\text{mode} = 60 + \frac{54 - 40}{2(54) - 40 - 34} \times 10 = 60 + \frac{140}{34} \approx 64.1$

Answer: mode $\approx$ 64 marks

2 Reading an ogive

2. An ogive for the monthly salaries of the workers in a factory passes through the points below. Find (a) the total number of workers (b) the median class (c) the lower-quartile class (d) the number of workers earning at least ₹6,000 but less than ₹10,000. **ICSE 2025, 4 marks**

Salary less than (₹)	2000	4000	6000	8000	10000	12000
Cumulative number of workers	10	20	35	55	70	75

SOLUTION

(a) The last cumulative frequency is the total. **75 workers.**

(b) $\frac{N}{2} = 37.5$. The cumulative frequency passes 37.5 between 6000 (35) and 8000 (55).

Median class: 6000–8000.

(c) $\frac{N}{4} = 18.75$. The cumulative frequency passes 18.75 between 2000 (10) and 4000 (20).

Lower-quartile class: 2000–4000.

(d) Workers below 10000 minus workers below 6000 = $70 - 35$. **35 workers.**

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